

Sinusoids

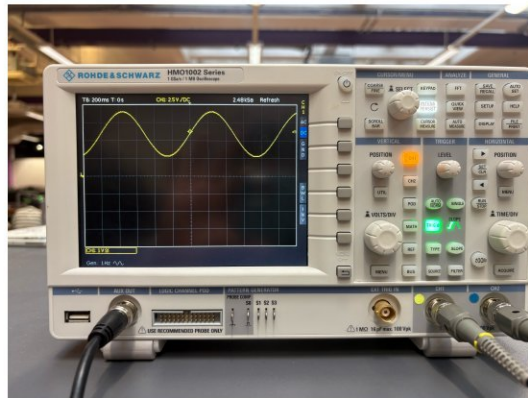
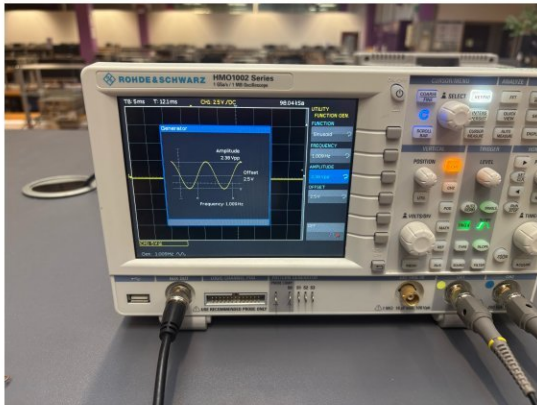
2025年10月21日 星期二 13:06

Sinusoids
14 October, 2025
Lab, Charles Wu

Reference: Lab skills. section 3.

Aim: Apply wave inputs to servos.

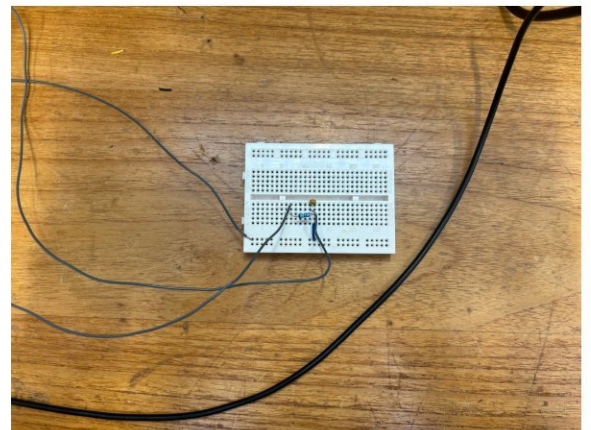
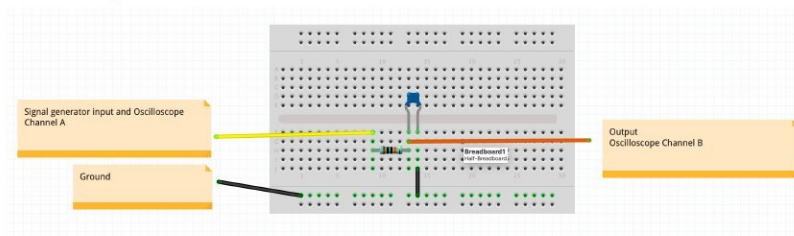
Procedure:
Generate a sin wave.



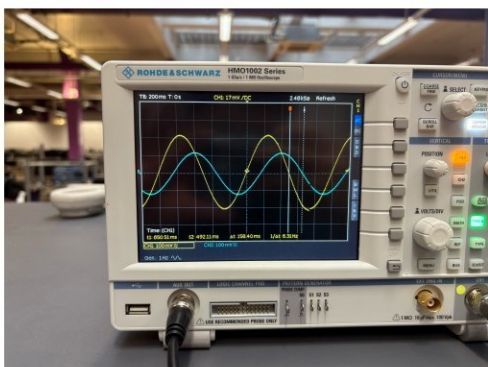
Servos oscillate.

[IMG_3411.MOV](#)

Making a phase shift.

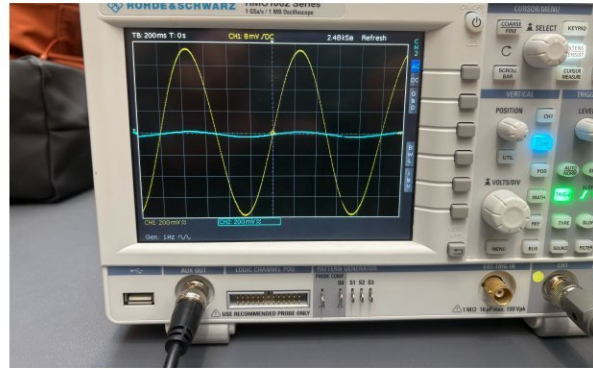
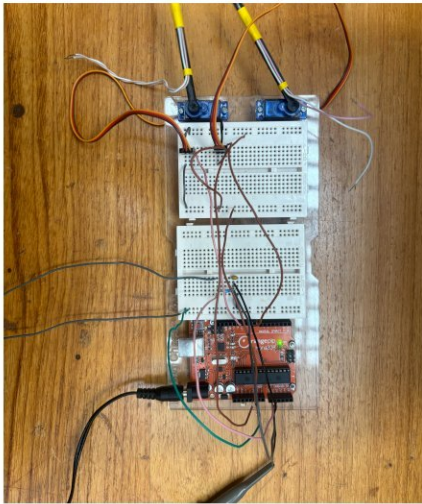


Circuit -



$$\text{Where phase difference} = \frac{2\pi t}{T} = \frac{2\pi \times 158.4}{1000} = 0.995$$



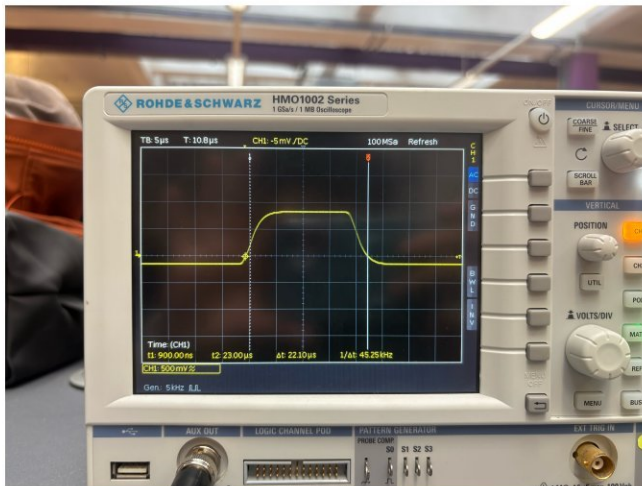


Servos moving out of phase.

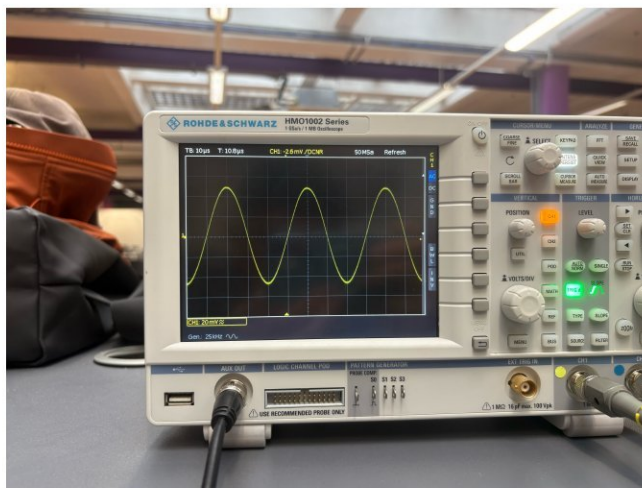
[IMG_3444.MOV](#)

Moving out of phase in different waves.

[IMG_3447.mov](#)



Pulse.



Sin.

AC channel only preserves AC signals.

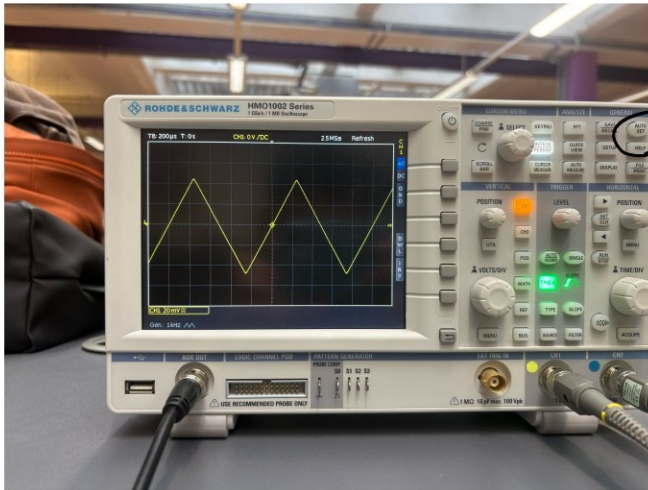
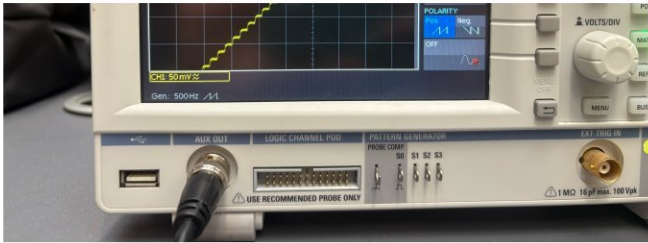
Wave is more centered and clearer.



Ramp.

Oscilloscope has limited sampling rate

So between time intervals, it creates steps.



Triangular
 → Autoset

是自动 -